



IMT Outbound On Premise API Doc V 1.4

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DEVELOPER API GUIDE DOCUMENT

PROJECT: Equity IMT Outbound On Premise API

Document

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1.0 TOKEN GENERATION

Method	POST
URL	https://10.11.6.111:8383/api/authenticate
Headers	Content-Type:application/json

1.0.1 Token Request Fields And Description

FIELD	DESCRIPTION	MANDATORY	TYPE	DEFAULT
password	password	YES	String	-
rememberMe		YES	boolean	true
username	username	YES	String	-

1.0.2 Token Sample Request

```
{  
  "password": "password",  
  "rememberMe": true,  
  "username": "username"  
}
```

1.0.3 Token Sample Response

```
{  
  "id_token":  
  "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbilzImF1dGgiOiJST0xFETUIOLFJPTEVfVVFuUilzImV4cCI6MTYxODExOTQxNX0.3DNUthKx-Z6OncvPgjh3MXsqlqyiwceUhy75btxel7OqbF_I2-b3xt_6N499XILZ38nnBLMDhwAxo-NuNM4DA"  
}
```

2.0 QUOTE GENERATION (C2C, C2B, And B2B)

Method	POST
URL	C2C: https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/c2c/quote C2B: https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/c2b/quote B2B: https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/b2b/quote
Headers	Content-Type:application/json

2.0.1 Quote Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	channel	Transacting channel Id	Y	String
2	referenceNumber	Unique identifier for the transaction	Y	String
3	rrn	Retrieval reference number (Can be same as referenceNumber)	Y	String
4	mode	Supply THUNES when sending to Thunes or MASTERCARD when sending to Mastercard	Y	String
5	transactionType	Transaction type - C2C, C2B, or B2B	Y	String
6	signature	Digital signature	Y	String
6	Source.bankId	Bank subsidiary: 11, 43, 50, 54, 55, or 56	N	String
7	source.currency	Source currency	Y	String
8	source.amount	Amount in KES	Y	String
9	source.accountNumber	Sender account number	Y	String
10	destination.countryCode	Destination country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
11	destination.bicWallet	1. For Mobile wallet, MOBILE_WALLET_ must be prefixed before the mobile wallet name (eg. MOBILE_WALLET_MTN Mobile Money). 2. For Bank transfer, destination.bicWallet holds the Swift code (bic code) for that particular bank (eg. BARCGB21XXX)	Y	String
12	destination.currency	Currency for the destination amount.	Y	String
13	destination.amount	Destination amount	Y	String

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14	destination.accountNumber	1. For bank transfer option, destination.accountNumber can either be an account number, iban, or clabe (Clabe is account number equivalent in Mexico) 2. For Mobile transfer option destination.accountNumber is the destination mobile number	Y	String
15	destination.sortCode	-	N	String
16	destination.bankAccountType	1. destination.bankAccountType list: CHECKING SAVINGS DEPOSIT OTHERS	N	String
17	destination.bankCode	1. destination.bankCode holds values for: i. Brazil corridor branch number ii. India corridor ifsCode ii. Australia corridor bsbNumber	N	String
18	destination.receivingBankName	Name of the receiving bank	N	String
19	receiver.firstName	First name of the receiver	Y	String
20	receiver.lastName	Last name of the receiver	Y	String
21	receiver.middleName	Middle name of the receiver	N	String
22	receiver.mobileNumber	Mobile number for the receiver	Y	String
23	receiver.email	Email address for the receiver	N	String
24	receiver.address.countryCode	Receiver country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
25	receiver.address.postalcode	Receiver postal code	Y	String
26	receiver.address.line1	Receiver address line	Y	String
27	receiver.address.line2	Receiver address line	N	String
28	receiver.address.city	Receiver city	Y	String

2.0.2 Quote Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	channel	Transacting channel Id	String
2	status	Quote status	String
3	responseCode	Response code	String
4	responseDesc	Response description	String
5	transactionId	Quote Id provided by partner. To be used during send transaction leg	String
6	rrn	Retrieval reference number	String
7	referenceNumber	Unique identifier for the transaction	String
8	mode	Transation mode	String
9	transactionType	Transaction type	String
10	signature	Digital signature	String
11	providerGL	Provider GL	String
12	payerName	Name of the payer provided by the partner	String
13	source.currency	Source currency	String
14	source.amount	Source amount	String
15	source.accountNumber	Source account number	String
16	destination.countryCode	Destination country code	String
17	destination.bicWallet	Destination bicWallet	String
18	destination.currency	Destination currency	String
19	destination.amount	Destination amount	String
20	destination.accountNumber	Destination account number	String
21	destination.sortCode	-	String
22	destination.bankAccountType	Destination account type	String
23	destination.bankCode	Destination bank code	String
24	destination.receivingBankName	Receiving bank name	String
25	charge.amount	Charge amount in KES	String
26	charge.currency	Charge currency - KES	String
27	thunesFees.amount	Thunes Fees	String
28	thunesFees.currency	Thunes Currency - USD	String

2.0.3 Quote Status Possible Values

STATUS(status)	DESCRIPTION
CREATED	Quote created successfully
FAILED	Failed to create quote

2.0.4 Sample C2C Quote Request

```
{
  "rrn": "8381728234",
  "referenceNumber": "2846528234",
  "mode": "THUNES",
  "transactionType": "C2C",
  "source": {
    "bankId": "54",
    "currency": "KES",
    "amount": "850875.000",
    "accountNumber": "0110100205683",
    "additionalProperties": {}
  },
  "destination": {
    "countryCode": "GBR",
    "bicWallet": "BOGGGB21XXX",
    "currency": "USD",
    "amount": "7500.0",
    "accountNumber": "DE05975683053163247100",
    "sortCode": "",
    "bankAccountType": "",
    "bankCode": "",
    "receivingBankName": "BOUSSARD AND GAVAUDAN GESTION",
    "additionalProperties": {}
  },
  "channel": "WEBB_APP",
  "signature":
    "pB1/m3f48KtsA1r13sDoIcYrgSb0zmhnNwFsbEHBaA6990HNaaNx8OxqQNahRPZQrAMkfW1Z6
    KgbOMiIVO1XZu3uYI5QB3zzRnivxlMAdMRU+rtbhmSxBgmbm/fM97+jSRjHOwSH5SQjBalZBY
    JW6hctHwMbmhHRp8CKXUXLHdmQXj95b4CzXjVskd1yez6hk7mynJJTnTL8otB25fPRnEho/TCjL
    qyF4UIR2anVsl8ss1s5eTmEj8gX/SAmWQCyHIwjjn97KRWhW3a92ZypLufwmo1c9n3bM1JMsBYp
    NZJSUtfJ5MPHzPgps/lnoyBtsJzB9Iggkqrx4FMdym0U4Q=",
  "receiver": {
    "firstName": "IMT",
    "lastName": "Test",
    "middleName": "Swift",
    "mobileNumber": "+442087599036",
    "email": "Linda.ashioya@equitybank.co.KE",
    "address": {

```

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```
    "countryCode": "GBR",
    "postalcode": "344544",
    "line1": "Terrible ",
    "line2": "",
    "city": "Aberdeen",
    "additionalProperties": {}
  },
  "additionalProperties": {}
},
"additionalProperties": {
  "sender": {
    "dateOfBirth": "1967-10-21",
    "nationality": "",
    "email": "MOBILE@TEST.COM",
    "firstName": "N/A",
    "lastName": "GICHUKI MUTHUI",
    "middleName": "",
    "mobileNumber": "25478890767",
    "address": {
      "city": "",
      "countryCode": "GBR",
      "line1": "",
      "line2": "",
      "postalcode": "10100"
    }
  }
}
}
```

2.0.5 Sample C2C Quote Response

```

{
  "rrn": "8381728234",
  "referenceNumber": "2846528234",
  "mode": "THUNES",
  "transactionType": "C2C",
  "source": {
    "currency": "KES",
    "amount": "7500.00",
    "accountNumber": "0110100205683",
    "additionalProperties": {}
  },
  "destination": {
    "countryCode": "GBR",
    "bicWallet": "BOGGGB21XXX",
    "currency": "USD",
    "amount": "7500.00",
    "accountNumber": "DE05975683053163247100",
    "sortCode": "",
    "bankAccountType": "",
    "bankCode": "",
    "receivingBankName": "BOUSSARD AND GAVAUDAN GESTION",
    "additionalProperties": {}
  },
  "status": "CREATED",
  "responseCode": "00",
  "responseDesc": "A quotation to All Banks / GBR / UK Faster Payment",
  "transactionId": "19214315",
  "channel": "WEBB_APP",
  "charge": {
    "amount": "1000.00",
    "currency": "KES",
    "additionalProperties": {}
  },
  "thunesFees": {
    "amount": "1.5",
    "currency": "USD",
    "additionalProperties": {}
  },
  "signature":
    "pB1/m3f48KtsA1r13sDoIcYrgSb0zmhnNwFsbEHBaA6990HNaaNx8OxqQNahRPZQrAMkfW1Z6
    KgbOMiIVO1XZu3uYI5QB3zzRnivxlMAdMRU+rtbhmSxBgmbm/fM97+jSRjHOwSH5SQjBalZBY
    JW6hctHwMbmhHRp8CKXUxLHdmQXj95b4CzXjVskd1yez6hk7mynJJTnTL8otB25fPRnEho/TCjL
    qyF4UIR2anVsl8ss1s5eTmEj8gX/SAmWQCyHIwjjn97KRWhW3a92ZypLufwmo1c9n3bM1JMsBYp
    NZJSUtfJ5MPHzPgps/InyoBtsJzB9Iggkqrx4FMdym0U4Q==",
  "providerGroup": "0541100405807",
  "payerName": "All Banks / GBR / UK Faster Payment",
  "additionalProperties": {}
}

```

}

3.0 C2C TRANSACTION POSTING

Method	POST
URL	https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/c2c/transaction
Headers	Content-Type:application/json

3.0.1 C2C Transaction Posting Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	channel	Transacting channel Id	Y	String
2	purposeofpayment	1. Purpose of payment can either be: "PERSONAL_TRANSFER", "SERVICE_CHARGE", "PROPERTY_PURCHASE", "PERSONAL_TRANSFER", "LOAN_PAYMENT", "TRAVEL", "GIFT_AND_DONATIONS", "FAMILY_SUPPORT", "UTILITY_BILLS", "EDUCATION", "MEDICAL_TREATMENT", "HOTEL_ACCOMMODATION", "INSURANCE_CLAIMS", "PROPERTY_RENTAL", "FUND_INVESTMENT", "SALARY_PAYMENT", "MAINTENANCE_EXPENSES", "SMALL_VALUE_REMITTANCE", "BUSINESS_INSURANCE", "SHARES_INVESTMENT", "OTHER"	Y	String
3	transactionId	Quotation Id received from the quotation response field, transactionId	Y	String
4	referenceNumber	Unique identifier for the transaction	Y	String
5	rrn	Retrieval reference number (Can be same as referenceNumber)	Y	String
6	mode	Supply THUNES when sending to Thunes or MASTERCARD when sending to Mastercard	Y	String
7	transactionType	Transaction type - C2C, C2B, or B2B	Y	String
8	destination.countryCode	Destination country code in ISO 3166-1 alpha-3 (3 digits) For Internal Use Only	Y	String

9	destination.bicWallet	1. For Mobile wallet, MOBILE_WALLET_ must be prefixed before the mobile wallet name (eg. MOBILE_WALLET_MTN Mobile Money). 2. For Bank transfer, destination.bicWallet holds the Swift code (bic code) for that particular bank (eg. BARCGB21XXX)	Y	String
10	destination.currency	Currency for the destination amount.	Y	String
11	destination.amount	Destination amount	Y	String
12	destination.accountNumber	1. For bank transfer option, destination.accountNumber can either be and account number, iban, or clabe(Clabe is account number equivalent in Mexico) 2. For Mobile transfer option destination.accountNumber is the destination mobile number	Y	String
13	destination.sortCode	-	N	String
14	destination.bankAccountType	1.destination.bankAccountType list: CHECKING SAVINGS DEPOSIT OTHERS	Y	String
15	destination.bankCode	1. destination.bankCode holds values for: i. Brazil corridor branch number ii. India corridor ifsCode ii. Australia corridor bsbNumber	N (Only mandatory for Brazil, India, and Australia)	String
16	destination.receivingBankName	Name of the receiving bank	N	String
16	Source.bankId	Bank subsidiary: 11, 43, 50, 54, 55, or 56	N	String
17	source.currency	Source currency	Y	String
18	source.amount	Amount in KES	Y	String
19	source.accountNumber	Sender account number	Y	String
20	sender.firstName	Sender first name	Y	String
21	sender.middleName	Sender Middle name	N	String
22	sender.lastName	Sender last name	Y	String
23	sender.mobileNumber	Sender mobile number	Y	String
24	sender.email	Sender email	N	String
25	sender.address.countryCode	Sender country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
26	sender.address.postalcode	Sender postalcode	Y	String
27	sender.address.line1	Sender address	Y	String
28	sender.address.line2	Sender address	N	String
29	sender.address.city	Sender city	Y	String
30	sender.dateOfBirth	Sender date of birth	Y	String
31	sender.nationality	Sender nationality	Y	String
32	receiver.firstName	First name of the receiver	Y	String
33	receiver.lastName	Last name of the receiver	Y	String

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34	receiver.middleName	Middle name of the reciver	N	String
35	receiver.mobileNumber	Mobile number for the receiver	Y	String
36	receiver.email	Email address for the receiver	N	String
37	receiver.address.countryCode	Receiver country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
38	receiver.address.postalcode	Receiver postal code	Y	String
39	receiver.address.line1	Receiver address line	Y	String
40	receiver.address.line2	Receiver address line	N	String
41	receiver.address.city	Receiver city	Y	String
42	orgReceiver.organizationName	This field holds the ID type of the beneficiary/ Receiver: "PASSPORT" "NATIONAL_ID" "DRIVING_LICENSE" "SOCIAL_SECURITY" "TAX_ID" "SENIOR_CITIZEN_ID" "BIRTH_CERTIFICATE" "VILLAGE_ELDER_ID" "RESIDENT_CARD" "ALIEN_REGISTRATION" "PAN_CARD" "VOTERS_ID" "HEALTH_CARD" "EMPLOYER_ID" "OTHER"	N (Only for corridors that require ID and ID type)	String
43	orgReceiver.registrationNumber	This field holds beneficiary/receiver ID number	N (Only for corridors that require ID and ID type)	String
44	orgReceiver.pin	Receiving organization's personal identification number	N	String
45	orgReceiver.mobileNumber	Receiving organization's mobile number	N	String
46	orgReceiver.email	Receiving organization's email	N	String
47	orgReceiver.address.countryCode	Receiving organization's country code	N	String
48	orgReceiver.address.postalcode	Receiving organization's postal code	N	String
49	orgReceiver.address.line1	Receiving organization's address	N	String
50	orgReceiver.address.line2	Receiving organization's address	N	String
51	orgReceiver.address.city	Receiving organization's city	N	String
52	signature	Digital signature	Y	String
53	channelCharges.amount	Bundled Charges from channel	N	String
54	channelCharges.currency	Charge currency	N	String

3.0.2 Transaction Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	rrn	Transacting channel Id	String
2	status	Transaction status	String
3	responseCode	Response code	String
4	responseDesc	Response description	String

3.0.3 Sample C2C Transaction Request

```
{
  "channel": "WEBB_APP",
  "purposeofpayment": "EDUCATION",
  "transactionId": "19381172",
  "rrn": "47080685224",
  "referenceNumber": "24529685224",
  "mode": "THUNES",
  "transactionType": "C2C",
  "source": {
    "bankId": "54",
    "currency": "KES",
    "amount": "1134.500",
    "accountNumber": "0220160435678",
    "additionalProperties": {}
  },
  "channelCharges": {
    "amount": "5",
    "currency": "USD",
    "additionalProperties": {}
  },
  "sender": {
    "firstName": "N/A",
    "middleName": "",
    "lastName": "MWANGI WATHIRA",
    "mobileNumber": "254763368230",
    "email": "INFOSECTEST@TEST.COM",
    "address": {
      "countryCode": "GBR",
      "postalcode": "20318",
      "line1": "",
      "line2": "",
      "city": "",
      "additionalProperties": {}
    },
    "dateOfBirth": "1976-01-01",
    "nationality": "kenyan",
    "additionalProperties": {}
  },
}
```

```

"destination": {
  "countryCode": "GBR",
  "bicWallet": "RIVEGB21XXX",
  "currency": "USD",
  "amount": "10.00",
  "accountNumber": "DE05975683053163247100",
  "sortCode": "",
  "bankAccountType": "",
  "bankCode": "",
  "receivingBankName": "1776 RIVERBANK LTD",
  "additionalProperties": {}
},
"receiver": {
  "firstName": "Name",
  "lastName": "UK",
  "middleName": "Name",
  "mobileNumber": "+ 4412345678",
  "email": "name@gmail.com",
  "address": {
    "countryCode": "GBR",
    "postalcode": "2855",
    "line1": "1199",
    "line2": "",
    "city": "Abbots Langley",
    "additionalProperties": {}
  },
  "additionalProperties": {}
},
"orgReceiver": {
  "organizationName": "",
  "registrationNumber": "",
  "pin": "",
  "mobileNumber": "",
  "email": "",
  "address": {
    "countryCode": "",
    "postalcode": "",
    "line1": "",
    "line2": "",
    "city": "",
    "additionalProperties": {}
  },
  "additionalProperties": {}
},
"signature":
"mhalmiUPrz4JbkyHwZYOTSeb1Zcc1nDJbzb9/VRiHxU2ksjx7NST1U3O+gK/bOPxRXSQT5P
BHmsbNUTC10x3XsiAeww2G5SLt2W2q/zmRzNTHrOlzT9/ZrCE0g/nVLvb3DbMpSUm3G5A
pP/11NJoU2GG1JxwWTkeXPq1ZcYjJn2L96rFLb+eJdkkE9kKKCvD8SucGOyR1sQqletYLqtk
JSDJAapDmVWwclRtcHPbM5H5XsiRnghOZhwjDCDzjsBsNZWAqKnEgQPTC7mQ2QO48dDe
LTPuYzE+1vavMfhkbJxf9+1+DvYB2Wcm3H89JJufwkplxSm0kyLKslZyJxN8Kzg=="
}

```


3.0.4 Sample C2C Transaction Response

```
{  
  "rrn": "47080685224",  
  "status": "POSTED",  
  "responseCode": "00",  
  "responseDesc": "COMPLETED"  
}
```

3.0.5 Transaction Status Possible Values

No	STATUS	RESPONSE CODE
1	POSTED	00
2	PENDING	001
3	FAILED	{Based on the error codes below}

4.0 C2B TRANSACTION POSTING

Method	POST
URL	https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/c2b/transaction
Headers	Content-Type:application/json

4.0.1 C2B Transaction Posting Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	channel	Transacting channel Id	Y	String
2	purposeofpayment	1. Purpose of payment can either be: "PERSONAL_TRANSFER", "SERVICE_CHARGE", "PROPERTY_PURCHASE", "PERSONAL_TRANSFER", "LOAN_PAYMENT", "TRAVEL", "GIFT_AND_DONATIONS", "FAMILY_SUPPORT", "UTILITY_BILLS", "EDUCATION", "MEDICAL_TREATMENT", "HOTEL_ACCOMMODATION", "INSURANCE_CLAIMS", "PROPERTY_RENTAL", "FUND_INVESTMENT", "SALARY_PAYMENT", "MAINTENANCE_EXPENSES", "SMALL_VALUE_REMITTANCE", "BUSINESS_INSURANCE", "SHARES_INVESTMENT", "OTHER"	Y	String
3	transactionId	Quotation Id received from the quotation response field, transactionId	Y	String
4	referenceNumber	Unique identifier for the transaction	Y	String
5	rrn	Retrieval reference number (Can be same as referenceNumber)	Y	String
6	mode	Supply THUNES when sending to Thunes or MASTERCARD when sending to Mastercard	Y	String
7	transactionType	Transaction type - C2C, C2B, or B2B	Y	String
8	destination.countryCode	Destination country code in ISO 3166-1 alpha-3 (3 digits)	Y	String

9	destination.bicWallet	1. For Mobile wallet, MOBILE_WALLET_ must be prefixed before the mobile wallet name (eg. MOBILE_WALLET_MTN Mobile Money). 2. For Bank transfer, destination.bicWallet holds the Swift code (bic code) for that particular bank (eg. BARCGB21XXX)	Y	String
10	destination.currency	Currency for the destination amount.	Y	String
11	destination.amount	Destination amount	Y	String
12	destination.accountNumber	1. For bank transfer option, destination.accountNumber can either be an account number, iban, or clabe (Clabe is account number equivalent in Mexico) 2. For Mobile transfer option destination.accountNumber is the destination mobile number	Y	String
13	destination.sortCode	-	N	String
14	destination.bankAccountType	1. destination.bankAccountType list: CHECKING SAVINGS DEPOSIT OTHERS	Y	String
15	destination.bankCode	1. destination.bankCode holds values for: i. Brazil corridor branch number ii. India corridor ifsCode ii. Australia corridor bsbNumber	N (Only mandatory for Brazil, India, and Australia)	String
16	destination.receivingBankName	Name of the receiving bank	N	String
16	Source.bankId	Bank subsidiary: 11, 43, 50, 54, 55, or 56	N	String
17	source.currency		Y	String
18	source.amount	Amount in KES	Y	String
19	source.accountNumber	Sender account number	Y	String
20	sender.firstName	Sender first name	Y	String
21	sender.middleName	Sender Middle name	N	String
22	sender.lastName	Sender last name	Y	String
23	sender.mobileNumber	Sender mobile number	Y	String
24	sender.email	Sender email	N	String
25	sender.address.countryCode	Sender country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
26	sender.address.postalcode	Sender postalcode	Y	String
27	sender.address.line1	Sender address	Y	String
28	sender.address.line2	Sender address	N	String
29	sender.address.city	Sender city	Y	String
30	sender.dateOfBirth	Sender date of birth	Y	String
31	sender.nationality	Sender nationality	Y	String
32	receiving_business.registered_name	Receiving business registered name	Y	String

33	receiving_business.trading_name	Receiving business trading name	Y	String
34	receiving_business.address	Receiving business address	Y	String
35	receiving_business.postal_code	Receiving business postal code	Y	String
36	receiving_business.city	Receiving business city	Y	String
37	receiving_business.province_state	Receiving business province	N	String
38	receiving_business.country_iso_code	Receiving business country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
39	receiving_business.msisdn	Receiving business mobile number	N	String
40	receiving_business.registration_number	Receiving business registration number	Y	String
41	receiving_business.taxId	Receiving business taxid	N	String
42	receiving_business.date_of_incorporation	Receiving business date of incorporation	N	String
43	receiving_business.representative_lastname	Receiving business representative last name	N	String
44	receiving_business.representative_lastname2	Receiving business representative last name	N	String
45	receiving_business.representative_firstname	Receiving business representative first name	N	String
46	receiving_business.representative_middlename	Receiving business representative middle name	N	String
47	receiving_business.representative_nativename	Receiving business representative native name	N	String
48	receiving_business.representative_id_type	Receiving business representative id type	N	String
49	receiving_business.representative_id_country_iso_code	Receiving business representative country code in ISO 3166-1 alpha-3 (3 digits)	N	String
50	receiving_business.representative_id_number	Receiving business representative id number	N	String
51	receiving_business.representative_id_delivery_date	Receiving business rep delivery date	N	String
52	receiving_business.representative_id_expiration_date	Receiving business rep expiration date	N	String
53	receiver.firstName	First name of the receiver	Y	String
54	receiver.lastName	Last name of the receiver	Y	String
55	receiver.middleName	Middle name of the receiver	N	String
56	receiver.mobileNumber	Mobile number for the receiver	Y	String
57	receiver.email	Email address for the receiver	N	String
58	receiver.address.countryCode	Receiver country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
59	receiver.address.postalcode	Receiver postal code	Y	String
60	receiver.address.line1	Receiver address line	Y	String
61	receiver.address.line2	Receiver address line	N	String

62	receiver.address.city	Receiver city	Y	String
63	orgReceiver.organizationName	This field holds the ID type of the beneficiary/Receiver: "PASSPORT" "NATIONAL_ID" "DRIVING_LICENSE" "SOCIAL_SECURITY" "TAX_ID" "SENIOR_CITIZEN_ID" "BIRTH_CERTIFICATE" "VILLAGE_ELDER_ID" "RESIDENT_CARD" "ALIEN_REGISTRATION" "PAN_CARD" "VOTERS_ID" "HEALTH_CARD" "EMPLOYER_ID" "OTHER"	N (Only for corridors that require ID and ID type)	String
64	orgReceiver.registrationNumber	This field holds beneficiary/receiver ID number	N (Only for corridors that require ID and ID type)	String
65	orgReceiver.pin	Receiving organization's personal identification number	N	String
66	orgReceiver.mobileNumber	Receiving organization's mobile number	N	String
67	orgReceiver.email	Receiving organization's email	N	String
68	orgReceiver.address.countryCode	Receiving organization's country code	N	String
69	orgReceiver.address.postalCode	Receiving organization's postal code	N	String
70	orgReceiver.address.line1	Receiving organization's address	N	String
71	orgReceiver.address.line2	Receiving organization's address	N	String
72	orgReceiver.address.city	Receiving organization's city	N	String
73	signature	Digital signature	Y	String

4.0.2 Transaction Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	rrn	Transacting channel Id	String
2	status	Transaction status	String
3	responseCode	Response code	String
4	responseDesc	Response description	String

4.0.3 Sample C2B Transaction Request

```
{
  "transactionId": "13591774",
  "rrn": "18947602008",
  "referenceNumber": "1237603008",
  "mode": "THUNES",
  "transactionType": "C2B",
  "source": {
    "bankId": "54",
    "currency": "KES",
    "amount": "7417.00",
    "accountNumber": "0570163595674",
    "additionalProperties": {}
  },
  "sender": {
    "firstName": "John",
    "middleName": "",
    "lastName": "Doe",
    "mobileNumber": "722000000",
    "email": "",
    "address": {
      "countryCode": "KES",
      "postalcode": "0100",
      "line1": "100",
      "line2": "",
      "city": "Nairobi",
      "additionalProperties": {}
    },
    "dateOfBirth": "1976-01-01",
    "nationality": "Kenyan",
    "additionalProperties": {}
  },
  "receiving_business": {
    "address": "100",
    "city": "London",
    "country_iso_code": "GBR",
    "date_of_incorporation": "",
    "email": "",
    "msisdn": "",
    "postal_code": "0100",

```

```

    "province_state": "",
    "registered_name": "XYZ Company Limited",
    "registration_number": "987650",
    "representative_firstname": "",
    "representative_id_country_iso_code": "",
    "representative_id_delivery_date": "",
    "representative_id_expiration_date": "",
    "representative_id_number": "",
    "representative_id_type": "",
    "representative_lastname": "",
    "representative_lastname2": "",
    "representative_middlename": "",
    "representative_nativename": "",
    "tax_id": "",
    "trading_name": "XYZ Company"
  },
  "destination": {
    "countryCode": "GBR",
    "bicWallet": "ABCEGB2L",
    "currency": "GBP",
    "amount": "64.30",
    "accountNumber": "DE05975683053163247100",
    "sortCode": "",
    "bankAccountType": "",
    "bankCode": "",
    "receivingBankName": "CITI",
    "additionalProperties": {}
  },
  "receiver": {
    "firstName": "John",
    "lastName": "Doe",
    "middleName": "",
    "mobileNumber": "0720000000",
    "email": "",
    "address": {
      "countryCode": "GBR",
      "postalcode": "100",
      "line1": "100",
      "line2": "",
      "city": "London",
      "additionalProperties": {}
    },
    "additionalProperties": {}
  },
  "orgReceiver": {
    "organizationName": "",
    "registrationNumber": "",
    "pin": "",
    "mobileNumber": "",
    "email": "",
    "address": {

```

```

    "countryCode": "",
    "postalcode": "",
    "line1": "",
    "line2": "",
    "city": "",
    "additionalProperties": {}
  },
  "additionalProperties": {}
},
"purposeofpayment": "EDUCATION",
"channel": "OTC",
"signature":
"2W2q/zmRzNTHrOIZT9/ZrCE0g/nVLvb3DbMsjx7NST1U3O+gK/bOPxRXSQTmhalmiUPrz4Jbky
HwZYOTSeb1Zcc1nDJbzb95PBHmsbNUTC10x3XsiAeww2G5SLtpSUM3G5ApP/rFLb+eJdkkE9kK
KCvD8SucGOyR1sQqltYLqtkJSBjAapDmMwcIRtcHPbM5H5XsiRnghOZhwjDCDzjsBsNZWAqK
nEgQPTC7mQ2QO48dDeLTPuYZE+1vavMfhkbJxf9+1+DvYB2Wcm3H89JJufwkplxSm0kyLKslZyj
xN8Kzg==",
  "additionalProperties": {}
}

```

4.0.4 Sample C2B Transaction Response

```

{
  "rrn": "18947602008",
  "status": "POSTED",
  "responseCode": "00",
  "responseDesc": "COMPLETED"
}

```

4.0.5 Transaction Status Possible Values

No	STATUS	RESPONSE CODE
1	POSTED	00
2	PENDING	001
3	FAILED	{Based on the error codes below}

5.0 B2B TRANSACTION POSTING

Method	POST
URL	https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/b2b/transaction
Headers	Content-Type:application/json

5.0.1 B2B Transaction Posting Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	channel	Transacting channel Id	Y	String
2	purposeofpayment	1. Purpose of payment can either be: "PERSONAL_TRANSFER", "SERVICE_CHARGE", "PROPERTY_PURCHASE", "PERSONAL_TRANSFER", "LOAN_PAYMENT", "TRAVEL", "GIFT_AND_DONATIONS", "FAMILY_SUPPORT", "UTILITY_BILLS", "EDUCATION", "MEDICAL_TREATMENT", "HOTEL_ACCOMMODATION", "INSURANCE_CLAIMS", "PROPERTY_RENTAL", "FUND_INVESTMENT", "SALARY_PAYMENT", "MAINTENANCE_EXPENSES", "SMALL_VALUE_REMITTANCE", "BUSINESS_INSURANCE", "SHARES_INVESTMENT", "OTHER"	Y	String
3	transactionId	Quotation Id received from the quotation response field, transactionId	Y	String
4	reference_number	Unique identifier for the transaction	Y	String
5	rrn	Retrieval reference number (Can be same as referenceNumber)	Y	String
6	mode	Supply THUNES when sending to Thunes or MASTERCARD when sending to Mastercard	Y	String
7	transaction_type	Transaction type - C2C, C2B, or B2B	Y	String
8	destination.countryCode	Destination country code in ISO 3166-1 alpha-3 (3 digits)	Y	String

For Internal Use Only

9	destination.bicWallet	1. For Mobile wallet, MOBILE_WALLET_ must be prefixed before the mobile wallet name (eg. MOBILE_WALLET_MTN Mobile Money). 2. For Bank transfer, destination.bicWallet holds the Swift code (bic code) for that particular bank (eg. BARCGB21XXX)	Y	String
10	destination.currency	Currency for the destination amount.	Y	String
11	destination.amount	Destination amount	Y	String
12	destination.accountNumber	1. For bank transfer option, destination.accountNumber can either be and account number, iban, or clabe (Clabe is account number equivalent in Mexico) 2. For Mobile transfer option destination.accountNumber is the destination mobile number	Y	String
13	destination.sortCode	-	N	String
14	destination.bankAccountType	1. destination.bankAccountType list: CHECKING SAVINGS DEPOSIT OTHERS	Y	String
15	destination.bankCode	1. destination.bankCode holds values for: i. Brazil corridor branch number ii. India corridor ifsCode ii. Australia corridor bsbNumber	N (Only mandatory for Brazil, India, and Australia)	String
16	destination.receivingBankName	Name of the receiving bank	N	String
16	Source.bankId	Bank subsidiary: 11, 43, 50, 54, 55, or 56	N	String
17	source.currency	Source	Y	String
18	source.amount	Amount in KES	Y	String
19	source.accountNumber	Sender account number	Y	String
20	sender.firstName	Sender first name	Y	String
21	sender.middleName	Sender Middle name	N	String
22	sender.lastName	Sender last name	Y	String
23	sender.mobileNumber	Sender mobile number	Y	String
24	sender.email	Sender email	N	String
25	sender.address.countryCode	Sender country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
26	sender.address.postalcode	Sender postalcode	Y	String
27	sender.address.line1	Sender address	Y	String
28	sender.address.line2	Sender address	N	String
29	sender.address.city	Sender city	Y	String
30	sender.dateOfBirth	Sender date of birth	Y	String
31	sender.nationality	Sender nationality	Y	String
32	receiving_business.registeredName	Receiving business registered name	Y	String

For Internal Use Only

33	receiving_business.tradingName	Receiving business trading name	Y	String
34	receiving_business.address	Receiving business address	Y	String
35	receiving_business.postalCode	Receiving business postal code	Y	String
36	receiving_business.city	Receiving business city	Y	String
37	receiving_business.provinceState	Receiving business province	N	String
38	receiving_business.countryIsoCode	Receiving business country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
39	receiving_business.msisdn	Receiving business mobile number	N	String
40	receiving_business.registrationNumber	Receiving business registration number	Y	String
41	receiving_business.taxId	Receiving business taxid	N	String
42	receiving_business.dateOfIncorporation	Receiving business date of incorporation	N	String
43	receiving_business.representativeLastName	Receiving business representative last name	N	String
44	receiving_business.representativeLastName2	Receiving business representative last name	N	String
45	receiving_business.representativeFirstName	Receiving business representative first name	N	String
46	receiving_business.representativeMiddleName	Receiving business representative middle name	N	String
47	receiving_business.representativeNativeName	Receiving business representative native name	N	String
48	receiving_business.representativeIdType	Receiving business representative id type	N	String
49	receiving_business.representativeIdCountryIsoCode	Receiving business representative country code in ISO 3166-1 alpha-3 (3 digits)	N	String
50	receiving_business.representativeIdNumber	Receiving business representative id number	N	String
51	receiving_business.representativeIdDeliveryDate	Receiving business rep delivery date	N	String
52	receiving_business.representativeIdExpirationDate	Receiving business rep expiration date	N	String
53	sending_business.registered_name	Sending business registered name	Y	String
54	sending_business.trading_name	Sending business trading name	Y	String
55	sending_business.address	Sending business address	Y	String
56	sending_business.postal_code	Sending business postal code	Y	String
57	sending_business.city	Sending business city	Y	String
58	sending_business.province_state	Sending business province	N	String
59	sending_business.country_iso_code	Sending business country code in ISO 3166-1 alpha-3 (3 digits)	Y	String
60	sending_business.msisdn	Sending business mobile number	N	String
61	sending_business.registration_number	Sending business registration number	Y	String

62	sending_business.taxId	Sending business taxid	N	String
63	sending_business.date_of_incorporation	Sending business date of incorporation	N	String
64	sending_business.representative_lastname	Sending business representative last name	N	String
65	sending_business.representative_lastname2	Sending business representative last name	N	String
66	sending_business.representative_firstname	Sending business representative first name	N	String
67	sending_business.representative_middlename	Sending business representative middle name	N	String
68	sending_business.representative_nativename	Sending business representative native name	N	String
69	sending_business.representative_id_type	Sending business representative id type	N	String
70	sending_business.representative_id_country_iso_code	Sending business representative country code in ISO 3166-1 alpha-3 (3 digits)	N	String
71	sending_business.representative_id_number	Sending business representative id number	N	String
72	sending_business.representative_id_delivery_date	Sending business rep delivery date	N	String
73	sending_business.representative_id_expiration_date	Sending business rep expiration date	N	String

5.0.2 Transaction Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	rrn	Transacting channel Id	String
2	status	Transaction status	String
3	responseCode	Response code	String
4	responseDesc	Response description	String

5.0.3 Sample B2B Transaction Request

```
{
  "channel": "OTC",
  "destination": {
    "accountNumber": "DE05975683053163247100",
    "amount": "64.30",
    "bankAccountType": "",
    "bankCode": "",
    "bicWallet": "ABCEGB2L",
    "countryCode": "GBR",
    "currency": "USD",
    "receivingBankName": "",
    "sortCode": ""
  },
  "mode": "THUNES",
  "purposeOfPayment": "OTHER",
  "receiving_business": {
    "address": "100",
    "city": "London",
    "country_iso_code": "GBR",
    "date_of_incorporation": "",
    "email": "",
    "msisdn": "",
    "postal_code": "0100",
    "province_state": "",
    "registered_name": "XYZ Company Limited",
    "registration_number": "987650",
    "representative_firstname": "",
    "representative_id_country_iso_code": "",
    "representative_id_delivery_date": "",
    "representative_id_expiration_date": "",
    "representative_id_number": "",
    "representative_id_type": "",
    "representative_lastname": "",
    "representative_lastname2": "",
    "representative_middlename": "",
    "representative_nativename": "",
    "tax_id": "",
    "trading_name": "XYZ Company "
  },
}
```



```

"reference_number": "68900100100108",
"rrn": "68900100100108",
"sender": {
  "address": {
    "city": "",
    "countryCode": "",
    "line1": "",
    "line2": "",
    "postalcode": ""
  },
  "dateOfBirth": "",
  "email": "",
  "firstName": "",
  "lastName": "",
  "middleName": "",
  "mobileNumber": "",
  "nationality": ""
},
"sending_business": {
  "address": "100",
  "city": "Nairobi",
  "country_iso_code": "KES",
  "date_of_incorporation": "",
  "email": "",
  "msisd": "",
  "postal_code": "",
  "province_state": "",
  "registered_name": "ABC Company Limited",
  "registration_number": "12345",
  "representative_firstname": "",
  "representative_id_country_iso_code": "",
  "representative_id_delivery_date": "",
  "representative_id_expiration_date": "",
  "representative_id_number": "",
  "representative_id_type": "",
  "representative_lastname": "",
  "representative_lastname2": "",
  "representative_middlename": "",
  "representative_nativename": "",
  "tax_id": "",
  "trading_name": "ABC Company"
},
"signature":
"XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX",
"source": {
  "accountNumber": "0570163595674",
  "amount": "7417.00",
  "currency": "KES"
},
"transactionId": "12729009",

```

```
"transaction_type": "B2B"  
}
```

5.0.4 Sample B2B Transaction Response

```
{  
  "rrn": "68900100100108",  
  "status": "POSTED",  
  "responseCode": "00",  
  "responseDesc": "COMPLETED"  
}
```

5.0.5 Transaction Status Possible Values

No	STATUS	RESPONSE CODE
1	POSTED	00
2	PENDING	001
3	FAILED	{Based on the error codes below}

6.0 TRANSACTION STATUS CHECK

Method	POST
URL	https://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/status/check
Headers	Content-Type:application/json

6.0.1 Transaction Status Check Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	channel	Transacting channel Id	Y	String
2	internalPaymentRef	Unique identifier for the transaction used as the referenceNumber field in the request	Y	String
3	paymentType	Transaction type - C2C, C2B, or B2B	Y	String
4	signature	Digital signature	Y	String
5	bankId	Source bank: 54 for KE, 43 for DRC	Y	String

6.0.2 Transaction Status Check Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	internalPaymentRef	Unique identifier for the transaction used as the referenceNumber field in the request	String
2	mode	THUNES	String
3	paymentType	Transaction type - C2C, C2B, or B2B	String
4	status	Transaction status	String
5	responseCode	Response code	String
6	responseDesc	Response description	String

6.0.3 Sample Transaction Status Check Request

```
{
  "channel": "OTC",
  "internalPaymentRef": "320032901104",
  "paymentType": "C2C",
  "bankId": "43",
  "signature":
  "j3se0HWHIIFVLsIAdl1k3N91wz4yqfKBvqiY/zAV59vB0whTNBWpPuH4jhviH3RxLSQnQM08W
  XWy06BzzTfiPqvN7b80IswL7O7SDz67vsk4qz3MsbNZ9u/UYaVy4GHMPjZY41Wob0NC1H3Hski
  umVWNkIvKUHMgfP3GmvlyAGJOfmafdeySzzDcTeZaBKLYtZQDMESaJvdQU/zxV00nWE
  /d+LAnF22Xj1mujV0X1nLWDe+jkjujd=="
}
```

6.0.4 Sample Transaction Status Check Response

```
{
  "internalPaymentRef": "320032901104",
  "mode": "THUNES",
  "paymentType": "C2C",
  "status": "SUCCESS",
  "responseCode": "00",
  "responseDesc": "COMPLETED"
}
```

6.0.5 Transaction Status Possible Values

No	STATUS	RESPONSE CODE
1	SUCCESS	00
2	PENDING	001
3	FAILED	{Based on the error codes below}

7.0 China Corridor Additional Requirements

7.0.1 Introduction

These additional requirements are applicable for China corridor through Thunes. Only two payers are allowed for China corridor:

- a. AliPay
- b. UnionPay

7.0.2 AliPay Corridor

7.0.2.1 Prerequisite

Sender confirms the exact name of the beneficiary, in pinyin, and beneficiary mobile number (MSISDN) with country code before the start of the transaction.

7.0.2.2 Quote Additional Steps

1. Beneficiary mobile number (MSISDN) with country code, is mapped to **destination.accountNumber** field.
2. BIC/ Swift Code value should always be **AliPay**. This value is mapped to the **destination.bicWallet** field.
3. Destination country code is **CHN**. This value is mapped to **destination.countryCode** field.
4. Destination currency is **CNY**. This value is mapped to **destination.currency** field.

7.0.2.3 Transaction Additional Steps

A. Receiver Details

1. Receiver name (in pinyin) should be mapped to `receiver.lastname` and `receiver.firstname` field
2. Beneficiary mobile number should be mapped to `receiver.mobileNumber` field

B. Sender Details

1. Sender firstname and lastname should be mapped to the `sender.firstName` and `sender.lastName` fields respectively.
2. Sender date of birth should be mapped to `sender.dateOfBirth` field
3. Sender mobile number should be mapped to `sender.mobileNumber` field
4. Sender id number should be mapped to `sender.email` field
5. Sender id type should be mapped to `sender.nationality` field
6. Sender address needs the following fields:
 - `sender.address.city`
 - `sender.address.line1`
 - `sender.address.postalcode`

7.0.3 UnionPay Corridor

7.0.3.1 Prerequisite

Sender confirms the exact name of the beneficiary in pinyin, Id number, and bank account number (bank account number = Union Pay Card Number) before the start of the transaction.

7.0.3.2 Quote Additional Steps

1. BIC/ Swift Code value should always be **UnionPay**. This value should be mapped to `destination.bicWallet` field
2. Beneficiary bank account number (Union Pay Card Number) should be mapped to **destination.accountNumber** field
3. Destination country code is **CHN**. This value is mapped to **destination.countryCode** field.
4. Destination currency is **CNY**. This value is mapped to **destination.currency** field.

7.0.3.2 Transaction Additional Steps

A. Receiver Details

1. Receiver name (in pinyin) should be mapped to `receiver.lastname` and `receiver.firstname` field
2. Beneficiary id number should be mapped to `orgReceiver.registrationNumber` field
3. Beneficiary bank account number (Union Pay Card Number) should be mapped to `destination.accountNumber` field
3. Beneficiary mobile number should be mapped to `receiver.mobileNumber` field

B. Sender Details

1. Sender firstname and lastname should be mapped to `sender.firstName` and `sender.lastName` fields respectively
2. Sender date of birth should be mapped to `sender.dateOfBirth` field
3. Sender mobile number should be mapped to `sender.mobileNumber` field
4. Sender id number should be mapped to `sender.email` field
5. Sender id type should be mapped to `sender.nationality` field

8.0 Response/ Error codes

8.0.1 List of Response/ Error Codes

1	Success		
	Code	Status	Description
	00	SUCCESS	Transaction Approved
	001	PENDING	Transaction Awaiting Final Status
2	Internal Related Errors		
	0001	FAILED	Input Validation Failure
	0002	FAILED	Authentication Failure
	0003	FAILED	Internal System Failure
	0004	FAILED	Duplicate
	0005	FAILED	Transfer Limit Failure
	0006	FAILED	Record Not Found
	0007	FAILED	Record Mismatch
	0008	FAILED	Message Format Error
	0009	FAILED	Invalid Request
	0010	FAILED	Transaction Not Permitted
	0999	FAILED	System Generic Error
3	Core Banking System - CBS Errors		
	1000	FAILED	CBS Get Service Failure
	1001	FAILED	CBS Transaction Posting Failure
	1002	FAILED	CBS Posting Timeout
	1003	FAILED	CBS Cannot Be Reached
	1004	FAILED	Invalid Account Number
	1005	FAILED	Insufficient Funds
	1006	FAILED	Cut over (end of day) in progress
	1999	FAILED	CBS Generic Error
4	HTTP Posting Errors		
	2000	FAILED	Http Request Posting Failure
	2001	FAILED	Http Request Timed Out
	2999	FAILED	Http Posting Generic Error
5	Third Party Errors		
	3000	FAILED	Failure In The Partner System
	3999	FAILED	Third Party System Generic Error
			For Internal Use Only

6	Anti Money Laundering - AML Errors		
	4000	FAILED	AML Decline
	4001	FAILED	AML Challenge
	4002	FAILED	AML Review
	4999	FAILED	AML Generic Error
7	Mobile Wallet Additional Errors		
	5000	FAILED	Mobile Wallet Validation Failure
	5001	FAILED	Unregistered Mobile Number
	5002	FAILED	Invalid Mobile Number
	5003	FAILED	Mobile Number In Unknown State
	5999	FAILED	Mobile Wallet Generic Error

9.0 Check Supported Corridor API

This API used to check whether a given country is supported by IMT

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/corridor/{country_code}
Headers	Authorization:Bearer {token}

9.0.1 Check Supported Corridors Field Description

Field	Description	Type	Mandatory
country_code	ISO 3166-1 alpha-3 country code	String	Y

9.0.2 Sample Check Supported Corridors Request

http://10.11.6.111:8383/services/imtcorems/api/outward/mc/v1/corridor/DEU

9.0.3 Sample Check Supported Corridors Response - Supported Corridors

```
{
  "supported":true,
  "corridors":[
    "MASTERCARD",
    "THUNES"
  ]
}
```

9.0.4 Sample Check Supported Corridors Response - Not Supported

```
{
  "supported":false,
  "providers":[]
}
```

10.0 Check Supported Thunes Services API

This API used to check which service type a given country is supports

Method	GET
URL	services/imtcorems/api/outward/mc/v1/service?{bank_id}&{country_iso_code}
Headers	Authorization:Bearer {token}

10.0.1 Check Supported Service Types Field Description

Url Param	Description	Type	Mandatory
country_iso_code	ISO 3166-1 alpha-3 country code	String	Y
bank_id	The subsidiary bank: 54 for Kenya 43 for DRC	String	Y

10.0.2 Sample Check Supported Service Types Request

`http://localhost:8383/services/imtcorems/api/outward/mc/v1/service?bank_id=54&country_iso_code=GHA`

10.0.3 Sample Check Supported Corridors Response - Supported Corridors

```
[
  {
    "id": 1,
    "name": "MobileWallet"
  },
  {
    "id": 2,
    "name": "BankAccount"
  }
]
```

10.0.4 Sample Check Supported Service Types Response – Invalid Country Code

Http Response Code: 400

```
{
  "id": 0,
```

```
"status": "FAILURE",  
"statusMessage": "Parameter 'country_iso_code' is invalid, country code expected, ISO 3166-1 alpha-3 format"  
}
```

10.0.4 Sample Check Supported Service Types Response – No Service Supported

```
[
  {
    "serviceId": 1,
    "countryIsoCode": "GHA",
    "bankId": 54,
    "payerId": 36,
    "payerName": "Airtel Money",
    "countryIsoCode": "GHA"
  }
]
```

11.0 Check Supported Thunes Payers Services API

This API used to check which payers are supported as per the service type of a given country

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/payers? {bank_id}{service_id}&{country_iso_code}
Headers	Authorization:Bearer {token}

11.0.1 Check Supported Service Types Field Description

Url Param	Description	Type	Mandatory
service_id	1 for MobileWallet 2 for BankAccount	int	Y
country_iso_code	ISO 3166-1 alpha-3 country code	String	Y
bank_id	The subsidiary bank: 54 for Kenya 43 for DRC	String	Y

11.0.2 Sample Check Supported Service Types Request

http://localhost:8383/services/imtcorems/api/outward/mc/v1/payers?bank_id=54&service_id=1&country_iso_code=GHA

11.0.3 Sample Check Supported Payer Response - Successful

```
[
  {
    "payerId": 36,
    "payerName": "Airtel Money",
    "countryIsoCode": "GHA"
  },
  {
    "payerId": 37,
    "payerName": "Airtel Money",
    "countryIsoCode": "GHA"
  }
]
```

```
[{"payerId": 37,
  "payerName": "MTN Mobile Money",
  "countryIsoCode": "GHA"},
{
  "payerId": 985,
  "payerName": "Tigo Money",
  "countryIsoCode": "GHA"},
{
  "payerId": 1399,
  "payerName": "Vodafone Money",
  "countryIsoCode": "GHA"}
]
```

11.0.4 Sample Check Supported Payers Response – Invalid Country Code

Http Response Code: 400

```
{
  "id": 0,
  "status": "FAILURE",
  "statusMessage": "Parameter 'country_iso_code' is invalid, country code expected, ISO 3166-1 alpha-3 format"
}
```

11.0.4 Sample Check Supported Service Types Response – No Payer Supported

```
[]
```

12.0 Check Thunes Payers Details – Mandatory Fields Services API

This API used to check payers mandatory fields as per the supplied service type, country code, and transaction type

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/payers/details? {bank_id}{service_id}&{payer_id}&{country_iso_code}&{transaction_type}
Headers	Authorization:Bearer {token}

12.0.1 Check Payer Mandatory Fields Description

Url Param	Description	Type	Mandatory
service_id	1 for MobileWallet 2 for BankAccount	int	Y
payer_id	payerId value as received from Thunes Payers Services API above	int	Y
country_iso_code	ISO 3166-1 alpha-3 country code	String	Y
transaction_type	Thunes supported transaction type: C2C C2B B2B	String	Y
bank_id	The subsidiary bank: 54 for Kenya 43 for DRC	String	Y

12.0.2 Sample Check Payer Mandatory Fields Request

http://localhost:8383/services/imtcorems/api/outward/mc/v1/payers/details?bank_id=54&service_id=2&payer_id=1243&country_iso_code=gha&transaction_type=C2C

12.0.3 Sample Payer Mandatory Fields Response – Successful for C2C

```
{
  "country_iso_code": "GHA",
  "currency": "GHS",
  "id": 1243,
  "increment": 0.0,
  "name": "Gcb Bank Ltd",
  "precision": 0,
  "service": {
    "id": 2,
    "name": "BankAccount"
  },
  "transaction_types": {
    "c2C": {
      "credit_party_identifiers_accepted": [
        [
          "bank_account_number"
        ]
      ],
      "credit_party_information": {
        "credit_party_identifiers_accepted": [
          [
            "bank_account_number"
          ]
        ]
      },
      "credit_party_verification": {
        "credit_party_identifiers_accepted": [
          [
            "bank_account_number"
          ]
        ]
      },
      "required_receiving_entity_fields": [
        []
      ]
    },
    "maximum_transaction_amount": "100000.00000000",
    "minimum_transaction_amount": "0.00000000",
    "purpose_of_remittance_values_accepted": [],
  }
}
```



```
"required_documents": [
  []
],
"required_receiving_entity_fields": [
  [
    "lastname",
    "firstname"
  ]
],
"required_sending_entity_fields": [
  [
    "lastname",
    "firstname",
    "date_of_birth",
    "country_iso_code"
  ]
]
},
"C2C": {
  "credit_party_identifiers_accepted": [
    [
      "bank_account_number"
    ]
  ],
  "credit_party_information": {
    "credit_party_identifiers_accepted": [
      [
        "bank_account_number"
      ]
    ]
  },
  "credit_party_verification": {
    "credit_party_identifiers_accepted": [
      [
        "bank_account_number"
      ]
    ]
  },
  "required_receiving_entity_fields": [
    []
  ]
}
```

```
    },
    "maximum_transaction_amount": "100000.00000000",
    "minimum_transaction_amount": "0.00000000",
    "purpose_of_remittance_values_accepted": [],
    "required_documents": [
      []
    ],
    "required_receiving_entity_fields": [
      [
        "lastname",
        "firstname"
      ]
    ],
    "required_sending_entity_fields": [
      [
        "lastname",
        "firstname",
        "date_of_birth",
        "country_iso_code"
      ]
    ]
  }
}
```

12.0.4 Sample Payer Mandatory Fields Response – Invalid Country Code

Http Response Code: 400

```
{
  "id": 0,
  "status": "FAILURE",
  "statusMessage": "Parameter 'country_iso_code' is invalid, country code expected, ISO 3166-1 alpha-3 format"
}
```

12.0.4 Sample Payer Mandatory Fields Response – Invalid Parameter

Http Response Code: 500

```
{
```

```
"id": 0,  
"status": "FAILURE",  
"statusMessage": "A Payer With Id: 1243 Does Not Exist In Country: gha For Transaction Type: C2B"  
}
```

13.0 Check Thunes ID Types API

This API used to A list of Thunes Accepted ID types

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/payers/id-types
Headers	Authorization:Bearer {token}

13.0.0 Sample Check Thunes ID Types Request

<http://localhost:8383/services/imtcorems/api/outward/mc/v1/id-types>

13.0.1 Sample Check Thunes ID Types Response

```
[
  {
    "type": "VOTERS_ID",
    "description": "Voter's Identification Card"
  },
  {
    "type": "HEALTH_CARD",
    "description": "Health Insurance Card/Number"
  },
  {
    "type": "EMPLOYER_ID",
    "description": "Employer Identification Card"
  },
  {
    "type": "OTHER",
    "description": "Others not listed"
  },
  {
    "type": "DRIVING_LICENSE",
    "description": "Driving License"
  },
  {
    "type": "VILLAGE_ELDER_ID",
```

```
    "description": "Village Elder Identification Card"
  },
  {
    "type": "NATIONAL_ID",
    "description": "National Identification Card"
  },
  {
    "type": "PAN_CARD",
    "description": "PAN Card"
  },
  {
    "type": "SOCIAL_SECURITY",
    "description": "Social Security Card/Number"
  },
  {
    "type": "ALIEN_REGISTRATION",
    "description": "Alien Registration Certificate/Card"
  },
  {
    "type": "RESIDENT_CARD",
    "description": "Permanent Residency Identification Card"
  },
  {
    "type": "TAX_ID",
    "description": "Tax Payer Identification Card/Number"
  },
  {
    "type": "SENIOR_CITIZEN_ID",
    "description": "Senior Citizen Identification Card"
  },
  {
    "type": "BIRTH_CERTIFICATE",
    "description": "Birth Certificate"
  },
  {
    "type": "PASSPORT",
    "description": "Passport"
  }
]
```


14.0 Check Thunes Purpose of Payment China Corridor API

This API used to A list of Thunes Accepted purpose of payment for China corridor.

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/payers/purpose/china
Headers	Authorization:Bearer {token}

14.0.0 Sample Url Request

<http://localhost:8383/services/imtcorems/api/outward/mc/v1/purpose/china>

14.0.1 Sample Response

```
[
  {
    "type": "FAMILY_SUPPORT",
    "description": "Family support"
  },
  {
    "type": "SALARY_PAYMENT",
    "description": "Payment of salary"
  },
  {
    "type": "ADVISORY_FEES",
    "description": "Fees for advisory or consulting service"
  },
  {
    "type": "LIBERALIZED_REMITTANCE",
    "description": "Liberalized remittance"
  },
  {
    "type": "PROPERTY_RENTAL",
    "description": "Property rental payment"
  },
  {
    "type": "ADVERTISING_EXPENSES",
```

```
    "description": "Advertising and/or public relations related expenses"
  },
  {
    "type": "CONSTRUCTION_EXPENSES",
    "description": "Construction expenses"
  },
  {
    "type": "EXPORTED_GOODS",
    "description": "Payments for exported goods"
  },
  {
    "type": "SERVICE_CHARGES",
    "description": "Payment for services"
  },
  {
    "type": "TAX_PAYMENT",
    "description": "Tax payment"
  }
]
```

15.0 Check Thunes Purpose of Payment- Other Corridors API

This API used to A list of Thunes Accepted purpose of payment for Other corridors.

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/payers/purpose
Headers	Authorization:Bearer {token}

15.0.0 Sample Url Request

http://localhost:8383/services/imtcorems/api/outward/mc/v1/purpose

15.0.1 Sample Response

Equity Bank Group

```
[
```



```
{
  "type": "SMALL_VALUE_REMITTANCE",
  "description": "Small value remittance"
},
{
  "type": "MEDICAL_TREATMENT",
  "description": "Medical treatment"
},
{
  "type": "PROPERTY_RENTAL",
  "description": "Property rental payment"
},
{
  "type": "EDUCATION",
  "description": "Education"
},
{
  "type": "ADVERTISING_EXPENSES",
  "description": "Advertising and/or public relations related expenses"
},
{
  "type": "FUND_INVESTMENT",
  "description": "Fund investment"
},
{
  "type": "BUSINESS_INSURANCE",
  "description": "Business related insurance payment"
},
{
  "type": "HOTEL_ACCOMMODATION",
  "description": "Hotel accommodation"
},
{
  "type": "LOAN_PAYMENT",
  "description": "Payment of loans"
},
{
  "type": "OFFICE_EXPENSES",
  "description": "Office expenses"
},
}
```

```
{
  "type": "ADVISORY_FEES",
  "description": "Fees for advisory or consulting service"
},
{
  "type": "OTHER_FEES",
  "description": "Broker, commitment, guarantee and other fees"
},
{
  "type": "UTILITY_BILLS",
  "description": "Utility bills"
},
{
  "type": "CONSTRUCTION_EXPENSES",
  "description": "Construction expenses"
},
{
  "type": "TAX_PAYMENT",
  "description": "Tax payment"
},
{
  "type": "FAMILY_SUPPORT",
  "description": "Family support"
},
{
  "type": "REWARD_PAYMENT",
  "description": "Payment of rewards"
},
{
  "type": "ROYALTY_FEES",
  "description": "Royalty, trademark, patent and copyright fees"
},
{
  "type": "PERSONAL_TRANSFER",
  "description": "Personal transfer"
},
{
  "type": "TRAVEL",
  "description": "Travel"
},
}
```

```
{
  "type": "LIBERALIZED_REMITTANCE",
  "description": "Liberalized remittance"
},
{
  "type": "EXPORTED_GOODS",
  "description": "Payments for exported goods"
},
{
  "type": "SERVICE_CHARGES",
  "description": "Payment for services"
},
{
  "type": "OTHER",
  "description": "Other purposes"
},
{
  "type": "GIFT_AND_DONATION",
  "description": "Gift and other donations"
},
{
  "type": "SALARY_PAYMENT",
  "description": "Payment of salary"
},
{
  "type": "DELIVERY_FEES",
  "description": "Delivery fees"
},
{
  "type": "PROPERTY_PURCHASE",
  "description": "Residential property purchase"
},
{
  "type": "TRANSPORTATION_FEES",
  "description": "Transportation fees"
},
{
  "type": "SHARES_INVESTMENT",
  "description": "Investment in shares"
},
}
```

```
{
  "type": "INFLUENCER_PAYMENT",
  "description": "Payment of Influencer"
},
{
  "type": "INSURANCE_CLAIMS",
  "description": "Insurance claims payment"
},
{
  "type": "COMPUTER_SERVICES",
  "description": "Computer service"
},
{
  "type": "MAINTENANCE_EXPENSES",
  "description": "Maintenance or other expenses"
}
]
```

16.0 Check Thunes Bank Types API

This API used to A list of Thunes Accepted bank types.

Method	GET
URL	/services/imtcorems/api/outward/mc/v1/payers/bank-types
Headers	Authorization:Bearer {token}

16.0.0 Sample Url Request

<http://localhost:8383/services/imtcorems/api/outward/mc/v1/bank-types>

16.0.1 Sample Response

```
[
  {
    "type": "OTHERS",
    "description": "Other type of account"
  },
  {
    "type": "SAVINGS",
    "description": "Savings Account"
  },
  {
    "type": "CHECKING",
    "description": "Checking Account"
  },
  {
    "type": "DEPOSIT",
    "description": "Deposit Account"
  }
]
```

17.0 Digital Signature Guide

For the purpose of ensuring data integrity and security, every channel will be required to generate a digital certificate and share the same with the maintainer of this API. The certificate will be used to verify the signature contained in each payload.

17.0.1 Generating The Signature

Each request should have a digital signature generated as described below:

1. Generate the clear text:

Concatenate the values in each request field shown below to generate the clear text. The clear text value will be hashed.

a. Clear Text For Quote

```
clearText = source.accountNumber  
          + source.amount  
          + source.currency  
          + destination.accountNumber  
          + destination.bicWallet  
          + destination.countryCode
```

b. Clear Text For Transaction

```
clearText = source.accountNumber  
          + source.amount  
          + source.currency  
          + destination.accountNumber  
          + destination.bicWallet  
          + destination.countryCode
```

b. Clear Text For Transaction Check Status

```
clearText = channel  
          + internalPaymentRef  
          + paymentType
```

17.0.2 Sign The Clear Text

a. Generate RSA private key

b. Using the private key and SHA-256 hashing algorithm, sign the clear text generated above.

In summary, the signature scheme should be SHA-256 with RSA

17.0.3 Sample Java Code

The figure below shows an example of how the signature can be generated using Java although the same can be achieved using any programming language

Example of how to generate a digital signature using Java

```
import java.security.PrivateKey;
import java.security.Signature;
import java.util.Base64;
public class SignMsg {
    private static final String UTF_8 = "UTF-8";
    private static final String ALGORITHM = "SHA256WithRSA";
    public String signMessage(String clearText , PrivateKey privatekey) {
        try {
            byte[] messageBytes = clearText .getBytes(UTF_8);
            Signature signatureInstance = Signature.getInstance(ALGORITHM);
            signatureInstance.initSign(privatekey);
            signatureInstance.update(messageBytes);
            byte[] digitalSignature = signatureInstance.sign();
            String signature = Base64.getEncoder().encodeToString(digitalSignature);
            return signature;
        } catch (Exception ex) {
            ex.printStackTrace();
            return null;
        }
    }
}
```